

Teacher's Solution Key: Assessment Problem 4

Solution & Marking Scheme (Total: 5 pts)

1. Coordinates of Particles (1.5 pts)

- **P3 Coordinates:** $x_3 = L \cos(53^\circ) = 10 \times 0.6 = \mathbf{6 \text{ cm}}$ [0.5 pt]
 $y_3 = L \sin(53^\circ) = 10 \times 0.8 = \mathbf{8 \text{ cm}} \implies P_3(6, 8)$ [0.5 pt]
- **P2 Coordinates:** Due to isosceles symmetry, the base length is $2 \times x_3 = 12 \text{ cm}$.
So, P_2 is at **(12, 0)**. [0.5 pt]

2. Total Mass (0.5 pts)

- **Answer:** $M_{sys} = 2 + 3 + 5 = \mathbf{10 \text{ kg}}$ [0.5 pt]

3. Center of Mass Coordinates (3 pts)

- **Abscissa x_G (1.5 pts):**

$$\begin{aligned}\sum m_i x_i &= (2)(0) + (3)(12) + (5)(6) \\ &= 0 + 36 + 30 = 66 && \text{(Correct moments) [0.5 pt]} \\ x_G &= \frac{66}{10} = \mathbf{6.6 \text{ cm}} && \text{(Final Result) [1.0 pt]}\end{aligned}$$

- **Ordinate y_G (1.5 pts):**

$$\begin{aligned}\sum m_i y_i &= (2)(0) + (3)(0) + (5)(8) \\ &= 0 + 0 + 40 = 40 && \text{(Correct moments) [0.5 pt]} \\ y_G &= \frac{40}{10} = \mathbf{4.0 \text{ cm}} && \text{(Final Result) [1.0 pt]}\end{aligned}$$

Final Result Result: The Center of Mass is located at $G(6.6, 4.0)$.